FishAI v0.5: Measuring Play Style Without Skill in a 54-Card Literature Variant

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August 2026

Abstract

FishAI v0.5 is a style-conditioned engine and simulation laboratory for Canadian Fish (Literature) under the `us54` rule dialect: 54 cards, nine half-suits of six, out-of-turn declares, any declaration error awarded to the opponents, and a race to five sets. Nine parameterized play styles share one identical full-strength inference engine, so that differences in outcome measure *style* rather than *skill*. The roster is evaluated in a full-precision duplicate-deal round-robin — 36 cells, 4,300 mirrored deal pairs per cell, 309,600 games, per-cell standard error at most 0.005 — and the payoff matrix is analyzed with mean score, maximin, Bradley–Terry, Nash averaging, α-Rank, and a Hodge decomposition into transitive and cyclic parts. The matrix is almost purely transitive (cyclic energy 0.0112 against a 0.15 threshold), and one style, *Punter*, satisfies all four pre-stated dominance criteria: highest mean score (0.5829), maximin above one half (0.5190), negligible cyclic energy, and exploitability 0.0025 against a rivals' median of 0.0444. Two measured caveats qualify the headline and are reported as first-class results. First, the declare-threshold axis that the roster's aggressive-to-conservative labels advertise is *inert* across the range the roster spans: the shared engine's confidence estimates are bimodal, so every threshold in the occupied band selects an identical set of declares, and Punter wins through its ask-scoring gamble bonus and uncertainty tolerance, not through a lower declare bar. Second, the card-hoarding style's signature benefit — a repeatable turn-pass through a team-contained book — was derived, implemented, and measured to be worth nothing: the Hoarder finishes 6th of 9 with and without it, and the best measured counter to hoarding is simply not hoarding. To the best of our knowledge, resting on an internal literature synthesis that found no peer-reviewed prior work on the game, this is the first quantitative game-AI study of Literature.

Addendum (August 2026). The tournament artifact every figure in this paper is read from has since been re-measured in place, against a roster carrying the concession layer and against a larger exploitability ladder. The verdict is unchanged — Punter is still dominant on all four criteria — but every headline figure above moved, and criterion 4 in particular is materially weaker. Section 11 reports the new numbers beside these, which are left exactly as published.

1 Introduction

Literature (Canadian Fish) is a six-player, two-team card game of public asks and hidden hands, in which every action is broadcast to the whole table and inference — not chance — decides who knows what. Communities that play it argue, persistently and without data, about temperament: whether the aggressive declarer beats the certainty purist, whether hoarding ask-licences is wisdom or superstition, whether the styles simply counter one another in a rock–paper–scissors loop. FishAI v0.5 was built to make that argument quantitative for one concretely pinned dialect. The research question, fixed before any tournament ran, is:

Is there a quantitatively superior play style — or do styles counter each other?

The question is harder to pose honestly than it looks, for two reasons. The first is a confound: a "style" implemented as a weaker bot measures nothing but its author's effort. FishAI's design rule is therefore that every style shares one identical, full-strength inference engine and differs only in the policy layer above it (Section 4). The second is that "best" is a meta-game notion, and meta-games are often intransitive: ranking systems that assume transitive skill, such as Elo, are actively misleading when the payoff matrix contains cycles [1]. The laboratory therefore decomposes the matrix into transitive and cyclic parts and commits, in advance, to a four-criterion decision rule under which "no dominant style; here is the counter-structure" is a first-class outcome, not a failure (Section 5).

This paper documents the v0.5 system and its measurements. Its contributions are:

1. **A rule-complete specification and engine for the `us54` dialect** — the 54-card "US student" variant — with its derived consequences: the sign flip in declaration risk, a tie-impossibility theorem, and the observation that whole-game termination is a property of the acting policy rather than of the rules (Section 3).

2. **A style-not-skill experimental design**: nine parameterized styles over one shared inference engine, a decision-divergence instrument that verifies each style is measurably distinct from the control, and duplicate-deal evaluation with shared seed sets and seat rotation (Sections 4–5).

3. **A full-precision tournament and its verdict**: 309,600 games over 36 cells, analyzed side by side with mean score, maximin, Bradley–Terry, Nash averaging, α-Rank and Hodge decomposition, yielding a *dominant* verdict for the Punter style under all four pre-stated criteria, together with the per-style diagnostics that explain the win (Section 6).

4. **Two negative results, reported as results.** The declare-threshold axis is inert across the roster's range, with a measured bimodal-confidence mechanism (Section 7); and the Hoarder style's thesis is refuted end-to-end — its costs measured, its theorized benefit implemented, and the benefit measured to be worth nothing (Section 8).

Throughout, every quantitative claim is traceable either to the committed tournament artifact [26] or to the repository's measurement documents [22, 23, 24, 25]; Section 5.7 records the artifact's provenance digests.

2 Related work

Literature as a game-AI domain. An internal research synthesis for this project [17] — a 65-style taxonomy of Literature play compiled with explicit evidence tiers — reports that targeted searches across arXiv, the ACM Digital Library, IEEE Xplore, AAAI proceedings, ResearchGate and university thesis repositories found *zero peer-reviewed papers, theses or technical reports on Literature as a game-AI domain*: no baseline agent, no published environment, no state-space or equilibrium analysis, no benchmark. That synthesis's own access to non-GitHub sources was constrained and it flags many of its secondary citations as unverified; we treat its absence claim as a careful search's negative report, current as of mid-2026, rather than as proof. The only public Literature reinforcement-learning project we are aware of is a four-player Q-learning implementation that reports no win rates [16]. The deepest strategy text on the game remains Develin's chapter [14], which independently attests the contained-book result used in Section 8.

Evaluating populations of agents. The evaluation stack is assembled from the multi-agent evaluation literature: Nash averaging, which plays a meta-game on the evaluation data itself and is robust to redundant entrants [1]; α-Rank, an evolutionary ranking that survives intransitivity [2]; and the Hodge-theoretic decomposition of a pairwise-comparison matrix into gradient (transitive) and cyclic parts [3, 1]. Bradley–Terry ratings [4] are reported alongside as the classical transitive fit. Duplicate (mirrored) deals are the card-game form of common random numbers, standard in duplicate bridge; the poker literature's stronger variance-reduction machinery [7] was measured to be unnecessary at this game's deal-dependence (Section 5.1). Sequential probability ratio tests [5] gate the tuning and exploitability searches, following modern chess-engine testing practice; the published matrix itself is fixed-N, since sequential stopping biases effect-size estimates. False-discovery control over the 36 simultaneous cells uses Benjamini–Hochberg [6].

Imperfect information and team play. Formally, Literature is a two-team constant-sum imperfect-information game whose teams may coordinate only ex ante — the regime whose solution concept is team-maxmin equilibrium with correlation [10]. All hidden information originates at the single chance node of the deal, and every subsequent event is public; the synthesis [17] notes that branching is tiny (at most 120 legal asks, typically 20–40) while one seat's deal-time information set in the 48-card baseline is on the order of 10^{24}, so belief fidelity, not search depth, is the game's difficulty. Search-based agents — ISMCTS [8] and perfect-information Monte Carlo [9] — are deliberately out of scope for v0.5: a search bot has a budget, not a style, and it would answer a different question. PIMC in particular must never evaluate declarations here, since strategy fusion makes every sampled world report the declaration succeeding [9, 25]. The style-vs-style setting also inherits the coordination literature's central warning: conventions formed in self-play need not transfer to novel partners [11, 12], which is why cross-play against an independently written bot is part of the design — and, not yet having been run, is listed as an open threat in Section 9. The shared inference engine itself belongs to the constraint-propagation family developed for deduction games such as Clue [13].

3 The game and the `us54` dialect

3.1 Base game

Six players in two teams of three, seated alternately, play with a deck partitioned into six-card *half-suits* ("sets" or "books"). On your turn you ask one specific card of one specific opponent; you must yourself hold a card of the asked set, and may not ask for a card you hold. A hit transfers the card face up and you ask again; a miss passes the turn to the player you asked. At any point a team may *declare* a set by naming the exact holder of all six cards, every assignment being a seat on the declarer's own team. Card counts and the full log of asks, results and declares are public. The team with more sets wins.

3.2 The `us54` dialect

FishAI v0.5 plays the "US student" dialect, specified in full with derived consequences and test vectors in the repository's rules document [22]. Table 1 summarizes the deltas against the 48-card baseline of the main reference rules text [15].

Out-of-turn declaring is resolved deterministically through a *declare window*: after every action, each seat in cyclic order from the turn-holder is offered the chance to declare; declining is itself a move, and when the turn-holder has no legal ask, declining is illegal (`MUST_DECLARE`) — a rule that

Table 1: The `us54` dialect against the 48-card baseline [22]. The 48-card wrong-declare outcome splits by error type (its rows 14–15): a set any opponent holds a card of is scored by the opponents — the same outcome as under `us54` — and only a set the declarer's own team wholly holds, misassigned, is void.

	`us54` (this work)	48-card baseline
Deck	54 — standard 52 plus two jokers	48 (8s removed)
Sets	9 half-suits of 6; the ninth is 8♣ 8♢ 8♡ 8♠ + both jokers	8
Hand	9 cards	8 cards
Declaring	any time, out of turn, in a declare window	own turn only
Wrong declare	**the opponents score the set**	opponent-held: they score; misassigned: void
Game end	the moment a team is awarded 5 sets	when all sets are resolved
Ties	impossible (Theorem 1)	possible (4–4)

Table 2: Declaration expected-value thresholds under the two rule sets [25].

Rule set	Outcome of any error	Declare iff
48-card baseline	misassigned: void; opponent-held: they score	$c > 1/(1+a)$
`us54`	opponents score (-1 us, $+1$ them)	$c \cdot a > 1/2$

guarantees a declare window can never stall [22]. Two of the dialect's rows change the strategy space far more than they appear to, and the next two subsections make them precise.

3.3 The sign flip in declaration risk

Under the 48-card baseline, misassigning a set your own team holds produces a *void*: zero for everyone. Under `us54` any error at all awards the set to the opponents. Relative to simply holding the set, a failed declaration used to cost one point (yours, burned); it now costs one point *and* hands the opponents one — a two-point swing in a race to five [23]. Writing c for the probability the set is contained by your team and a for the probability your seat-assignment is exactly right given containment, the expected-value thresholds differ as in Table 2 [25].

Every declaration threshold tuned against the 48-card game is therefore too loose under `us54`. The roster's thresholds are treated as 48-card *appetites* and mapped through $t_{\mathrm{us54}} = (1 + t_{\mathrm{pagat48}})/2$ before shipping — Blitz 0.70 → 0.85, Punter 0.55 → 0.775, Hoarder 0.95 → 0.975 [23]. Section 7 reports the measured, and unexpected, fate of this entire axis.

3.4 Ties are impossible — and termination is a policy property

Theorem 1 (Tie impossibility) *Under `us54`, no final state is level: every game that terminates ends in a clinch at exactly five sets, and a tie can never be emitted.*

Proof. Because the void outcome is abolished, every resolved set is awarded to exactly one team. There are nine sets. If neither team has reached five, each holds at most four, so at most eight sets are resolved — hence if all nine were resolved, some team reached five first, a contradiction with the game still running. The clinch target $\lfloor 9/2 \rfloor + 1 = 5$ is therefore always reached before the sets run out, and no final state can be level [22]. □

The qualifier in the theorem is deliberate. The pigeonhole argument guarantees that declares cannot run out before a clinch; it does not force declares to happen. Literature can *livelock*: an adversarial ask policy that always takes the last legal ask ping-pongs misses between two seats forever, with the position repeating exactly — no rule violated, every state holding a legal action. This reproduces under both rule sets and is a property of the game, not the dialect [22]. The architectural consequence is that whole-game termination belongs to the acting policy: the simulator's step cap (5,000) is load-bearing and instrumented, and in the reported tournament it never fired (Section 5.7). The engine itself is pure, deterministic TypeScript — the same seed reproduces a byte-identical game — and adversarially audited: a 250-game lockstep differential against the pre-variant engine (8,155 reduce steps, zero state mismatches), 130,480 deliberately illegal actions confirming stable error codes, and a 10,000-game `us54` fuzz gate asserting that every game terminates in a clinch at exactly five sets with no tie ever emitted [21].

4 Measuring style without skill

4.1 The design rule

The central confound in any style comparison is that a "style" may simply be a weaker bot. FishAI's rule: **every style shares one identical, full-strength inference engine** — a constraint-propagation knowledge layer over the public log — and styles differ only in the parameterized policy above it [24]. Bots consume only `SeatView`, the exact public-information payload a human in that seat would see, and a test enforces that no other hand is reachable: cheating is structurally impossible rather than policed [21]. One guardrail is imposed on every style: the certainty bonus in ask scoring stays at or above its baseline, because below it a style can rank an uncertain ask above a certain hit — which is not a style but a bug that would dominate the results [23].

4.2 The parameter surface

A style is a `StyleParams` vector [23] over six groups of knobs:

- *Ask scoring*: weights on hit probability (`wHit`), progress toward nearly-secured sets (`wProgress`), information gain (`wNarrow`); a floor on acceptable hit probability (`minHitP`); a bonus for asks that would complete a set (`gambleBonus`); and the protected certainty bonus.
- *Declare policy*: the confidence bar (`declareThreshold`), tolerated guessed cards (`declareMaxUncertain`), and the certainty-only and own-hand-only extremes.
- *Declare window* (new under `us54`): how early in the window to fire (`declareEagerness`); whether and at what separate bar to declare sets the seat holds no card of (`foreignDeclare`, `foreignDeclareThreshold`).
- *Clinch* (new under `us54`): preference for the closing declare at four sets (`clinchAggression`) and weight on denying the opponents' fifth (`denialWeight`).
- *Information and targeting*: the leak-protection tiebreak (`leakEpsilon`, `leakThreshold`), signalling asks, and whom to promote on a likely miss (`missTarget`).
- *Hoarding and the turn-pass*: keep ask-licences in at least N sets (`hoardBooks`), refuse declares that shrink the hand below N (`minHandSize`), and the contained-book turn-pass appetite (`containedPass`, Section 8.3).

Table 3: The nine-style roster. Settings are the shipped `us54` values; declaration thresholds are 48-card appetites mapped through $(1+t)/2$ (Section 3.3).

Style	Family	Thesis	Defining settings (shipped)
Balanced	control	the tuned `us54` baseline every other style is read against	baseline weights; `declareThreshold` 0.90; `containedPass` 1
Blitz	aggressive	tempo and sets now; information is cheap	`wHit` 90, `wProgress` 30, `declareThreshold` 0.85, `declareEagerness` 0.9, `leakEpsilon` 0, no signalling, `missTarget most`
Punter	aggressive	chase the completing card; accept the gift risk	`gambleBonus` 25, `minHitP` 0, `declareThreshold` 0.775, `declareMaxUncertain` 2
Banker	conservative	never gift a set; bank only certainties	`declareOnlyWhenCertain`, `minHitP` 0.25, `declareEagerness` 0.2
Turtle	passive	minimum risk; declare only sets wholly in hand	`declareOnlyOwnHand`, `minHitP` 0.4, no signalling, no foreign declares
Hoarder	optionality	keep ask-licences, stay alive, delay	`hoardBooks` 3, `minHandSize` 2, `declareThreshold` 0.975, `declareEagerness` 0.1, `containedPass` 1.33
Scout	information	deduce first, collect later	`wNarrow` 40, `wHit` 55, `declareOnlyWhenCertain`
Ghost	information	deny opponents the read	`leakEpsilon` 6, `leakThreshold` 3, no signalling
Archivist	information	track sets you hold nothing in; declare them for teammates	`foreignDeclare` at threshold 0.95, `wNarrow` 30, `declareEagerness` 0.7

4.3 The roster

Nine styles (Table 3). Their full shipped vectors are embedded in the committed artifact [26]; the table lists each style's defining deviations from the Balanced control.

Two roster notes matter for reading everything that follows. First, the family labels narrate an aggressive-to-conservative spectrum of *declaring*; Section 7 shows by measurement that the axis those labels advertise does not fire anywhere in the roster's range, so the labels must be read as naming the styles' *appetites*, not their realized declaring behaviour. The styles are nonetheless measurably distinct (Section 7.1) — through their ask policies, their gating extremes, and their window and hoarding knobs. Second, the Archivist is the style `us54` created: out-of-turn declaring makes it worthwhile to spend inference on sets one can never ask into, purely to declare them for teammates.

5 Methodology

5.1 Duplicate deals, shared seeds, seat rotation

One trial is one seeded deal played in both orientations — the style assignment swapped between the teams, the start seat held identical — so that a lucky deal cancels within the pair, exactly as in duplicate bridge; this is the common-random-numbers estimator. Measured on this engine at design time (500 mirrored pairs), pairing reduced the variance of the score estimate by a factor of 1.36 relative to independent games [24]; the realized per-cell variance ratios in the reported tournament average 1.33 (range 1.14–2.23) [26]. This is far below the 30× reductions achievable in poker with AIVAT-style control variates [7] — expected, because the whole deck is dealt and outcomes are far less deal-dominated — and the measured gain is why no heavier machinery was built. Every cell uses the same seed list (prefix `style-v1`), so cross-cell comparisons share deals too; start seats rotate $i \bmod 6$; and because the policies are deterministic, sample diversity comes entirely from seeds — the harness asserts `distinctSeeds` = pairs.

5.2 Precision and primary metric

The published matrix is fixed-N at 4,300 pairs (8,600 games) per cell — above the 2,600-pair budget derived for a target standard error of 0.005 — and the largest realized cell SE is exactly 0.0050 [26]. The primary metric is score rate (win = 1, tie = $\frac{1}{2}$, loss = 0), adopted because the 48-card game produces ties in roughly a quarter of mirror games [24]. Under `us54` ties are impossible (Theorem 1), so score rate coincides with win rate; the artifact asserts zero ties across all 309,600 games rather than rendering a column that cannot populate. Sequential tests are used only where they belong: SPRT ($\alpha = \beta = 0.05$) gates tuning and exploitability-search candidates [5], while the reported matrix is fixed-N because sequential stopping biases effect-size estimates [24].

5.3 Diagnostics

Alongside the payoff matrix, the lab records per-style diagnostics explaining *why* a style wins or loses: ask hit rate, turn retention, declare precision and yield, and a set of `us54`-specific metrics [23]: `concedeRate` (declares that gifted the set — the successor of the 48-card void rate, measuring a different event), `foreignDeclareRate` (declares of sets the declarer held no card of), `declareLatency` (window-cycles between a set becoming provable and the team declaring it, separating "knew and waited" from "never knew"), `raceLosses` (a teammate declared wrongly a set this seat could have declared correctly), and `clinchDenials`. One shipped metric is explicitly excluded from style comparisons: `hoardIndex`, a time-averaged count of sets held, does *not* measure hoarding — a hoarding style lengthens games, adding late small-hand samples that drag the mean down, a confound measured to be large enough to invert the sign [23].

5.4 The verdict rule

The decision rule was fixed in the experimental-design document before the tournament ran [24]: declare a superior style only if *all four* hold —

1. highest mean score; and
2. maximin > 0.5 (no losing matchup); and
3. cyclic energy of the matrix below 0.15; and
4. exploitability not materially worse than its rivals'.

If (2) or (3) fails, the published answer is "no dominant style; here is the counter-graph and the equilibrium mixture" — rendered as a first-class result. Cyclic energy is the cyclic fraction of the Hodge decomposition of the duplicate-averaged antisymmetric payoff matrix [3, 1]; the equilibrium mixture is the maximum-entropy Nash of that matrix [1]; α-Rank supplies the evolutionary ranking [2]. With 36 simultaneous cells, roughly 1.8 would appear significant at $\alpha = 0.05$ by chance alone, so every significance label — including the individual 3-cycles behind criterion 3 — is Benjamini–Hochberg corrected [6].

5.5 The decision-divergence instrument

Distinctness is not assumed from parameters; it is measured at the decision level. Two policy vectors are compared by replaying real `us54` games on the lab's own seed set and, at *every* decision point, handing both vectors the identical `SeatView` and seed and comparing their two chosen `GameActions` [23]. The instrument returns the exact fraction of decisions on which the vectors differ; zero means the vectors are the same player. It is used three ways below: to verify each style differs from the control (Section 7.1), to expose the inert threshold axis (Section 7), and to detect that the first Hoarder implementation was byte-identical to Balanced (Section 8).

5.6 Exploitability search

For each style i, a best-response search over `StyleParams` deviations from i's own vector — candidates drawn from a fixed knob ladder, each SPRT-gated, with an accepted candidate re-measured by a fixed-N 400-pair evaluation on seeds no search decision saw — reports $E(i) = \mathrm{SR}(\text{best response}, i) - 0.5$ [24, 23]. E distinguishes "champion of this roster" from "actually strong": a style can top the table yet be maximally exploitable by a style nobody has written.

5.7 Health gates and provenance

A run is void unless its health counters are clean. For the reported tournament [26]: `illegalActions` 0, `cappedGames` 0, `invariantViolations` 0, `ties` 0, `voids` 0, `nonClinch` 0, `distinctSeeds` 4,300/4,300. The artifact is committed at `src/lab/data/style-results.v2.json` with records digest `0c3cb524a3534d4a`, engine commit `1667a1d` (recorded `1667a1d-dirty`, the suffix reflecting generation from a modified working tree at that commit), and the SHA-256 of the rules document (`b4c7fef7...`) so that results become detectably stale if the rules move. The full run — 309,600 games plus analysis — took roughly 16 minutes of wall clock. Two matrix artifacts exist: *v1* (engine commit `819eebb`, likewise recorded `819eebb-dirty`), and *v2*, re-run on the same seed set after the contained-book turn-pass was implemented (Section 8.3); all headline numbers in this paper are v2, with v1 cited where the paired comparison is itself the result.

Addendum. The path `src/lab/data/style-results.v2.json` now holds a *third* artifact: matrix v2 re-measured at engine commit `1fdd22e`, records digest `1d7d9f30b59c00f6` [27]. The artifact this paper reports — digest `0c3cb524a3534d4a` — is at that path in the repository history, at the commit preceding the re-measurement. Section 11 reports what moved; v1 is untouched.

6 Results

6.1 The verdict: dominant — Punter

All four criteria of Section 5.4 hold, and the artifact's computed verdict is *dominant* (Table 4). This is the criteria mechanism working as designed, not a narrative choice: the same pipeline renders "cyclic" or "inconclusive" when the numbers say so.

Addendum. All four still hold on the re-measured artifact, and the computed verdict is still *dominant* — but all four measured values in Table 4 moved, and criterion 4 moved a long way. Section 11 prints the new column beside this one.

Table 4: The four dominance criteria, as computed in the committed artifact [26].

#	Criterion	Measured (matrix v2)	Pass
1	highest mean score	Punter 0.5829, ahead of Blitz 0.5602 (margin 0.0227)	yes
2	maximin > 0.5	0.5190 (worst matchup: Balanced); CI_{95} lower bound 0.5119; worst cell significant after BH	yes
3	cyclic energy < 0.15	0.0112; zero significant 3-cycles after BH	yes
4	exploitability not materially worse	E(Punter) = 0.0025 vs. rivals' median 0.0444	yes

Table 5: Ranking systems side by side (matrix v2, 4,300 pairs per cell) [26]. B–T is the Bradley–Terry rating in Elo-like units.

Style	Mean score	95% CI	Maximin	Worst vs.	Nash mass	B–T	α-Rank
Punter	0.5829	[0.5798, 0.5860]	0.5190	Balanced	1.000	52.2	1
Blitz	0.5602	[0.5570, 0.5633]	0.4802	Punter	0.000	37.8	2
Balanced	0.5581	[0.5551, 0.5612]	0.4810	Punter	0.000	36.5	3
Banker	0.5285	[0.5252, 0.5318]	0.4531	Punter	0.000	17.9	4
Ghost	0.4928	[0.4894, 0.4962]	0.4133	Punter	0.000	−4.4	5
Hoarder	0.4885	[0.4854, 0.4917]	0.4205	Punter	0.000	−7.1	7
Archivist	0.4802	[0.4769, 0.4835]	0.4074	Punter	0.000	−12.3	6
Scout	0.4063	[0.4030, 0.4096]	0.3397	Punter	0.000	−59.1	9
Turtle	0.4025	[0.3992, 0.4058]	0.3419	Punter	0.000	−61.5	8

6.2 The ranking, five ways

Table 5 shows the five ranking systems side by side. They agree almost everywhere — which is itself evidence of transitivity. The mean-score, Bradley–Terry and Hodge-rating orderings are identical position for position; α-Rank differs only in swapping two adjacent pairs deep in the table: Hoarder/Archivist (mean-score gap 0.0083, about 3.6 combined standard errors of the means, though the pair's head-to-head cell is itself a statistical tie) and Scout/Turtle (gap 0.0038, about 1.6 combined standard errors — a statistical tie in mean score). Nash averaging places all its mass on Punter, and α-Rank concentrates on it likewise — the degenerate mixtures that a transitive matrix with a strict maximin winner forces.

The worst-vs. column carries a structural fact: every style's worst matchup is Punter. Punter itself has no losing matchup; its narrowest, against Balanced and Blitz, are two cells that are statistically tied with each other (0.5190 and 0.5198) — indeed, between matrix v1 and v2 the identity of Punter's worst opponent flipped between these two, a re-ordering of tied cells rather than a new counter [23]. Punter's full row is Table 6: every one of its eight cells is above 0.5 and significant after BH correction.

6.3 Matrix structure: almost no cycles

Thirty-three of the 36 cells are significant after BH correction. The three that are not are exactly the near-ties one would hope a well-powered design exposes as near-ties: Balanced–Blitz (0.4962, $q = 0.33$), Hoarder–Archivist (0.5074, $q = 0.12$) and Ghost–Archivist (0.4944, $q = 0.26$). The Hodge decomposition puts the cyclic energy at 0.0112 — about 1.1% of a matrix whose verdict threshold is 15%. Exactly one directed 3-cycle exists in point estimates (Hoarder > Archivist > Ghost > Hoarder, minimum edge 0.5056), and it is not significant after BH ($q = 0.27$) [23, 26]. The counter-structure that the laboratory was built to detect, had it existed, is measurably absent at

Table 6: Punter's matchup row (score from Punter's side; all cells BH-significant) [26].

Opponent	Punter's score	SE
Balanced	0.5190	0.0036
Blitz	0.5198	0.0042
Banker	0.5469	0.0048
Hoarder	0.5795	0.0042
Ghost	0.5867	0.0049
Archivist	0.5926	0.0047
Turtle	0.6581	0.0046
Scout	0.6603	0.0046

this precision: the styles form a ladder, not a wheel.

6.4 Why Punter wins

Table 7 pools each style's diagnostics over its eight cells. Punter's profile is volume at near-control precision: it makes the most declares per game (4.08 against the control's 3.99) and converts the most sets (4.11 against 4.02), while its declare precision gives up only 0.2 points against the control (0.9884 vs. 0.9901) and its concession rate rises only from 0.0099 to 0.0116. It pays measurable costs — more chosen concessions (0.0066 of its declares against the control's 0.0042) and more race losses (0.045 per game against 0.037) — and the two-point swing of Section 3.3 prices those costs; they are simply smaller than the value of the extra conversions.

The mechanism behind that profile is *not* the one the style's label advertises. Section 7 shows that Punter's low declare threshold (0.775) selects exactly the same declares as the control's 0.90: the axis is inert. What differs is the ask policy — `gambleBonus` 25 steers Punter toward asks that would complete a set, and `declareMaxUncertain` 2 lets it tolerate a second guessed card where every other style tolerates at most one. Punter wins as the style that hunts completions and accepts the priced risk of the gift rule, not as the style that declares on thinner evidence.

At the other end, the table explains the floor. Turtle's own-hand-only gate leaves it holding sets it cannot bank; its declares are forced ones over a third of the time (0.367 against the control's 0.069), its precision collapses to 0.9456, its concession rate is five times the control's, and it converts barely 2.94 sets per game. Scout buys information with its ask weights (`wHit` 55, `wNarrow` 40) and pays for it in material: the second-lowest hit rate and the fewest conversions of the non-Turtle styles. Ghost is the mirror image — the roster's *highest* hit rate (0.6025), bought by leak-protection that steers its asks away from informative sets — and still finishes below 0.5: under `us54`, protecting the read is worth less than the tempo it costs. The Hoarder's line is read in Section 8.

6.5 Exploitability

Table 8 reports $E(i)$ for each style with the accepted best-response deviation. Punter's E of 0.0025 — its best measured counter being merely to randomize the miss target — confirms it is not a paper champion of this particular roster. The ordering tracks the ranking: the exploitable styles are the ones already at the bottom, with Turtle's single-knob repair (turn off the own-hand-only gate) worth a full 0.15.

Two honest cautions accompany the table [23]. E is a maximum over a stochastic search whose final evaluations carry standard errors of 0.008–0.017 and which only accepts moves worth at least $\approx$0.03, so small entries and small differences between runs are not meaningful. And the search's knob ladder predates the contained-book turn-pass, so that one axis was probed separately by the

Table 7: Per-style diagnostics, pooled over each style's eight cells (matrix v2) [26]. "Hit" = ask hit rate; "Prec" = declare precision; "Conc" = concession rate (share of the style's declares gifted to the opponents); "Frgn" = foreign-declare rate; "Lat" = declare latency (window-cycles); "Race" = race losses per game; "Sets" and "Decl" = sets converted and declares made per game; "Forced" = share of declares that were compelled.

Style	Hit	Prec	Conc	Frgn	Lat	Race	Sets	Decl	Forced
Punter	0.564	0.988	0.012	0.013	22.8	0.045	4.11	4.08	0.066
Blitz	0.564	0.990	0.010	0.014	22.9	0.036	4.03	4.00	0.067
Balanced	0.568	0.990	0.010	0.014	23.2	0.037	4.02	3.99	0.069
Banker	0.563	0.990	0.010	0.014	22.8	0.035	3.93	3.90	0.072
Ghost	0.603	0.989	0.011	0.010	24.3	0.035	3.77	3.74	0.099
Hoarder	0.545	0.975	0.025	0.253	30.6	0.088	3.66	3.69	0.148
Archivist	0.556	0.967	0.033	0.007	19.8	0.091	3.76	3.79	0.082
Scout	0.532	0.957	0.043	0.007	17.2	0.088	3.49	3.52	0.080
Turtle	0.523	0.946	0.054	0.060	63.4	0.162	2.94	3.00	0.367

Table 8: Exploitability, matrix v2: $E(i)$ and the accepted best-response deviation from the style's own vector [26, 23].

Style	E	Accepted deviation
Balanced	0.0000	none (mirror)
Banker	0.0012	`gambleBonus` 15
Punter	0.0025	`missTarget random`
Blitz	0.0175	`leakEpsilon` 3
Archivist	0.0325	`wHit` 100
Hoarder	0.0563	`hoardBooks` 0, `minHandSize` 0 — *stop hoarding*
Ghost	0.0775	`wProgress` 30, `gambleBonus` 30
Scout	0.1112	`wHit` 90, `wNarrow` 0, `minHitP` 0.45
Turtle	0.1500	`declareOnlyOwnHand` off

same SPRT protocol: no candidate was accepted for any style at any appetite — consistent with Section 8.3's finding that the mechanism is worth nothing.

Addendum. The second caution has since been acted on rather than merely stated: the ladder now carries `containedPass` outright, along with two knobs that did not exist when this table was measured, and the truncation named below in Section 11.1 was removed. Every number in Table 8 is superseded, and Punter's is superseded worst of all — 0.0025 becomes 0.0525. Section 11.4 is the accounting.

7 Caveat I: the declare-threshold axis is inert

This section states the finding, its mechanism, and its consequences for reading the results; the full treatment — instrument, attribution, repairs, and the methodological argument — is a companion paper published alongside this one [18].

7.1 The styles are distinct. . .

Run through the decision-divergence instrument (Section 5.5) against the Balanced control, every non-control style differs on a measurable fraction of decisions [23]:

Table 9: Decision divergence when only `declareThreshold` varies on the Balanced control (40 games) [23]. All roster styles ship thresholds in $[0.775, 0.98]$.

`declareThreshold`	0.50	0.65	0.70	0.75	**0.775**	0.85	0.90	0.95	0.975
divergent decisions	350	48	29	19	**0**	0	0	0	0
% of decisions	1.297	0.178	0.108	0.070	**0.000**	0.000	0.000	0.000	0.000

	Scout	Ghost	Archivist	Banker	Turtle	Punter	Hoarder	Blitz
% of decisions differing	2.89	2.15	1.66	1.19	0.83	0.47	0.47	0.39

Between 0.39% and 2.89% of decisions may look small, but Literature is a game where a single redirected ask changes who holds the turn and what the whole table learns; the tournament shows these divergence rates compounding into mean-score gaps of up to 18 points.

7.2 . . . but not along the axis their labels advertise

`declareThreshold` is the knob on which the roster's aggressive-to-conservative story is built. Varying it *alone* on the Balanced control, over 40 `us54` games with every decision compared under the instrument, gives Table 9: the divergence collapses to exactly zero at 0.775 and stays zero at every measured point through 0.975 (the pilot sweep's zero extends through threshold 1) — and *every roster style's shipped threshold sits at* 0.775 *or above*, the shipped span being $[0.775, 0.98]$. The axis is dead for all nine of them [23].

The mechanism is not an unreachable code path — Balanced makes 171 speculative declares to 138 certain ones over those same 40 games, so the thresholded branch fires constantly. (An earlier audit, described in Section 8.1, had read a related zero as the branch never being reached; the direct sweep corrected that reading.) The measured mechanism is that the shared inference engine's confidence estimates are **bimodal**: a candidate declaration is either well above $\sim$0.975 or well below $\sim$0.775, with almost nothing in between, so any threshold inside that band selects the identical set of declares [23].

Three consequences follow, and each is stated wherever the ranking is published. First, Punter did not win by declaring at 0.775 rather than 0.90; it won through `gambleBonus` and `declareMaxUncertain` (Section 6.4). Second, any description of this roster as a spectrum from aggressive to conservative *declaring* describes a knob that does not fire; the labels survive only as names for appetites. Third, the $(1 + t)/2$ appetite mapping of Section 3.3, while arithmetically right, is currently unfalsifiable on this engine: every value it can produce in the plausible range yields identical play [23].

8 Caveat II: the Hoarder — costs measured, benefit implemented, still worthless

The Hoarder is the project's originally named style, and it is mechanically grounded: holding at least one card of a set is the *only* licence to ask into it, so declaring a set revokes your own asking rights there. Its story across v0.5 is the paper's second negative result, and it runs in three acts: an implementation that turned out to be the control in disguise; an audited implementation that expresses the thesis and loses by it; and the style's theorized benefit, implemented and measured to be worth nothing. The third act's mechanism — the contained-book turn-pass, its theorem, trigger, and measurements — is treated in full in a second companion paper published alongside this one [19].

8.1 Act one: the inert implementation

In the 36-cell pilot (200 pairs per cell), the Hoarder's mean score was 0.5484 — identical to Balanced's in every digit — its rows of the payoff and diagnostics tables were byte-identical to the control's, and the head-to-head cell measured 0.5000 with a standard error of exactly zero. Rather than infer, the identity was instrumented: over 400 games and 275,380 decision points, the Hoarder and Balanced vectors, handed identical views and seeds, chose *zero* different actions [23].

Attributing the zero one parameter at a time (each of the Hoarder's seven behavioural deltas applied to Balanced alone, over 76,972 decisions) returned zero for all seven. The audit's first finding was structural: the two hoard knobs gated only the speculative-declare branch, on the argument that refusing *free* points to protect an ask-licence is a giveaway rather than optionality — but within the measured population that branch never selected differently across the whole threshold band (the flatness later explained by Section 7's bimodality), so the gate guarded a distinction without a difference, and with it the entire style. The fix moved the hoard gate in front of *every* declare the style is free to refuse — speculative, certain, or wholly in hand — while never touching the compulsory `MUST_DECLARE` positions. The two knob values were then derived from the rule set rather than tuned: `minHandSize` 2 is the smallest hand that survives one involuntary strip (hits are compulsory) and still holds a licence, the only hand-size discontinuity in the rules being at zero cards; `hoardBooks` 3 is the style's stated appetite, corrected to count only licences with a legal ask behind them (a set the seat would hold all six of grants a licence with nothing legal to name) [23].

8.2 Act two: expressible, and wrong

After the audit the same instrument measures 9,315 divergent decisions in 289,560 (3.22%): the style exists. Every diagnostic moves in the direction the thesis predicts (Table 10): it stays in the asking game longer (dropout rate $0.521 \rightarrow 0.443$), waits 35% longer to bank (declare latency $22.90 \rightarrow 31.02$ window-cycles), and — because the sets it does declare are disproportionately the ones that cost it no licences — its foreign-declare rate rises thirteen-fold, the project owner's "memorize half-suits you do not own" strategy arriving as a side effect of hoarding. And it pays for all of it: the head-to-head against Balanced moves from 0.5000 ± 0.0000 to 0.5775 ± 0.0159 *against* the Hoarder, and its pilot mean falls from 0.5484 to 0.4738. Ablated on identical deals, `hoardBooks` 3 alone reproduces the entire effect (0.4225 to the Hoarder, identical to both knobs together, because any declare leaving under two cards also leaves under three licences); with both knobs off the vector returns to 0.5000 ± 0.0000 — the inertness finding restated as an outcome [23].

In the full-precision matrices the audited Hoarder finishes 6th of 9 (the pilot's 0.4738 figure corresponds to a 7th-place pilot finish; both full-precision runs place it 6th [23]). At this point an objection remained open, and the project's containment analysis [25] had sharpened it: the verdict was measured on an implementation that paid hoarding's full costs while collecting none of its benefit.

8.3 Act three: the benefit, implemented — the contained-book turn-pass

Call a set *contained* when all six of its cards sit with one team. Containment is an absorbing state with measured consequences [25]: no opponent can legally ask into the set (asking requires holding a card of it); an opponent declaring it necessarily assigns to their own team and thus gifts it straight back; a holder may still ask into it — a legal, *guaranteed* miss that passes the turn to a chosen

Table 10: The Hoarder before and after the audit — pilot diagnostics pooled over its eight cells [23]. The inert version is byte-identical to the control. Two Balanced figures coexist in the source: 0.5484 is the pooled value in the identity comparison that flagged the inert Hoarder, while this table's Balanced column pools Balanced's own eight cells (0.5581). The load-bearing identity is byte-identical play — zero divergent decisions in 275,380 — and the table reports both figures rather than resolving what the source does not.

Diagnostic (pilot)	Balanced	Hoarder, inert	Hoarder, audited
mean score	0.5581	0.5484 (= Balanced)	0.4738
foreign-declare rate	0.018	0.018	0.247
declare latency	23.39	22.90	31.02
race losses / game	0.044	0.046	0.091
declares / game	4.01	3.98	3.67
forced-declare share	0.074	0.076	0.153
concession rate	0.011	0.012	0.025
dropout rate	0.539	0.521	0.443
ask hit rate	0.568	0.569	0.545

opponent; the miss moves no cards, so the licence is never consumed; and declaring the set destroys the move. Claiming a contained set therefore has no defensive value, and leaving it unclaimed preserves a repeatable, aimable turn-pass — advice independently attested by Develin's strategy text [14]. This is precisely the benefit a hoarder's unclaimed sets should be earning.

v0.5 implements the turn-pass as a policy option with a derived, unturned trigger. Writing p^* for the hit probability of the ask the style would otherwise play, the pass is taken iff

$$p^* \;<\; \frac{a \cdot \text{gain} - \text{infoCost}}{\text{tempo}},$$

with every term in units of cards and read off the public log: the expected turn length $E = \text{hits}/\max(1, \text{misses})$ prices a conceded turn at $E \cdot n_t/\bar{n}$ for an opponent holding n_t cards (mean live hand $\bar{n}$), gain is the spread between the opponent the ordinary ask would concede to and the opponent the style's `missTarget` prefers, $\text{tempo} = 1 + E \cdot n_o/\bar{n}$ is the swing between hitting and missing, and infoCost is zero on a reuse of an already-published card and $1/U$ otherwise (U the count of live cards not yet publicly placed) [23]. The appetite a is the expected number of uses before the set is banked: 1 for eight of the nine styles, and 1.33 for the Hoarder — the one style whose delay had actually been measured, the number read off its own declare-latency ratio. A certain hit is never displaced, and the policy reuses one card, so 94.8–97.6% of uses by the heavy users cost no information at all. One honest check on the derivation: measured uses per contained book order the styles exactly as the appetite argument predicts, but the Hoarder-to-Balanced ratio comes out 1.18, not the 1.33 the latency ratio implied — right in sign, roughly right in size, and deliberately *not* re-fitted, since tuning an appetite to the behaviour it produces would be fitting to the outcome [23].

Table 11 reports what the mechanism does. Three readings. First, the opportunity is a property of the *declare* policy: a contained set exists at 54% of the Turtle's ask decisions (it never banks sets split across teammates) and 0.4% of the Scout's (it pins holders early, and a pinned contained set is banked immediately) — and the Hoarder sits second at 14.95%, nearly four times the control's 3.89%, which is exactly the benefit the objection said it was paying for and not collecting. Second, even where the licence exists it is mostly refused (3.8–16% taken), always on the derivation's gain ≤ 0 arm: the ordinary ask was already conceding the turn where the style wanted it. Blitz needs no exception — its `missTarget most` makes the aiming gain non-positive, so the mechanism turns itself

Table 11: The contained-book turn-pass: opportunity, firing, and value [23]. Opportunity and firing rates are from 100 mirror games per style; the score is the style *with* the mechanism against the identical style without it, over 1,200 duplicate pairs (2,400 games); "changed" counts games the mechanism actually touched.

Style	Appetite	Opportunity %	Fired / opp. %	Divergence %	Score (with vs. without)	Changed
Balanced	1	3.89	7.7	0.041	0.5046 ± 0.0034	239
Blitz	1	5.05	0.0	0.000	0.5000 ± 0.0000	0
Punter	1	3.65	3.8	0.019	0.4971 ± 0.0029	142
Banker	1	5.04	13.9	0.098	0.5021 ± 0.0035	281
Turtle	1	53.86	7.3	0.558	0.5008 ± 0.0065	1,268
Hoarder	1.33	14.95	12.4	0.258	0.4938 ± 0.0054	915
Scout	1	0.38	0.0	0.000	0.5000 ± 0.0000	0
Ghost	1	1.31	16.0	0.029	0.4958 ± 0.0027	156
Archivist	1	1.01	4.8	0.007	0.4996 ± 0.0004	11

Table 12: Matrix v1 vs. v2: per-style mean score, paired over the same 4,300 deals [23]. Every delta is within 1.5 SE of zero; every rank is unchanged.

Style	v1	v2	Δ	SE(Δ)	t	Rank
Punter	0.5829	0.5829	−0.0001	0.0009	−0.07	$1 \to 1$
Blitz	0.5594	0.5602	+0.0008	0.0008	+0.94	$2 \to 2$
Balanced	0.5584	0.5581	−0.0003	0.0009	−0.34	$3 \to 3$
Banker	0.5274	0.5285	+0.0011	0.0011	+1.00	$4 \to 4$
Ghost	0.4928	0.4928	+0.0000	0.0010	+0.03	$5 \to 5$
Hoarder	0.4904	0.4885	−0.0018	0.0014	−1.35	$6 \to 6$
Archivist	0.4791	0.4802	+0.0011	0.0008	+1.47	$7 \to 7$
Scout	0.4064	0.4063	−0.0001	0.0007	−0.12	$8 \to 8$
Turtle	0.4032	0.4025	−0.0007	0.0019	−0.38	$9 \to 9$

off. Third, and decisively: **no style is two standard errors from** 0.5. The largest deviation is 1.6 SE (Ghost), the two heavy users land on and slightly below the null (Turtle 0.5008 ± 0.0065, Hoarder 0.4938 ± 0.0054), and the styles that never fire return exactly 0.5000 ± 0.0000 over 2,400 games each — the with- and without-vectors are the same player. *The measured value of the contained-book turn-pass, as a policy option under* `us54` *against this roster, is zero.* The arithmetic says why: with measured $E \approx 1$, the whole spread between the best and worst opponent to concede to is on the order of one card, against which the displaced ask was worth p^* of a card plus a retained turn. The move is real, and it is small.

8.4 The verdict on the strategy

The full 36-cell matrix was then re-run with the mechanism live, at the same precision on the same seed set, so v1 and v2 differ by the policy change and nothing else. Nothing that matters moved (Table 12): the verdict, the complete ranking, the Nash and α-Rank concentrations, and every rank are identical; no cell moved significantly (the largest paired per-deal difference, Blitz–Ghost at $+0.0030 \pm 0.0015$, $t = 2.07$, does not survive BH over 36 comparisons — one $|t| > 2$ in 36 is what chance produces); cyclic energy went $0.0097 \to 0.0112$ with the same single non-significant 3-cycle; and the significant-cell count's $34 \to 33$ change is entirely the Hoarder–Archivist cell drifting out of significance (q $0.036 \to 0.118$) [23].

For the Hoarder specifically, the prediction that the missing benefit explained its finish *did* *not hold.* Its paired delta is -0.0018 ± 0.0014 ($t = -1.35$) — the point estimate went *down*, indistinguishably from zero; not one of its eight cells moved significantly, and its one marginal cell moved against significance. The sharpest form of the result comes from the exploitability search, free to pick any single knob: the best response to the Hoarder-with-the-benefit is `hoardBooks` 0, `minHandSize` 0 — *stop hoarding* — worth 0.5563 ± 0.0124 (Table 8). One number from the follow-up single-axis search deserves recording precisely because half of it would have looked like a finding: probing `containedPass` = 0 against the Hoarder, the SPRT search seeds said the candidate was *worse* by 0.020 ± 0.018 while the held-out evaluation said *better* by 0.0275 ± 0.0098; two contradictory effects of that size mean neither is real, and the honest summary is noise [23].

Addendum. The paired delta above stands: it is a statement about the two artifacts it compares, both of which are unchanged as measurements. The *sharpest* form, though, does not survive the re-measurement. On the larger ladder the accepted best response to the Hoarder is no longer `hoardBooks` 0, `minHandSize` 0 but `minHitP` 0.45, `leakEpsilon` 0 — and that candidate did not replicate: it scored 0.54125 in search and 0.480 ± 0.0125 on the held-out block, so E(Hoarder) clamps to 0.0000 [27]. Read that as a search that found nothing it could reproduce, *not* as the Hoarder having become unexploitable, and read "stop hoarding" as a best response of the old ladder rather than a standing result. The Hoarder also finishes 7th of 9 on the new matrix rather than 6th (Table 14). What the re-measurement does not touch is the finding this section is about — the turn-pass is worth nothing — which rests on the mirror measurement of Table 11 and the paired v1/v2 comparison of Table 12, neither of which reads the re-measured artifact.

So the conclusion is about the strategy, not the implementation. The benefit the containment analysis identified is real, reachable, and fires for the derived reasons — and it is worth nothing at this roster's level of play. The verdict stands with no unimplemented benefit behind it as an excuse: **optionality under `us54` is real, purchasable, and priced above its worth**, and the two rules doing the pricing are the ones Section 3.3 flagged — the gift rule makes a teammate's wrong declare of your hoarded set a two-point event (the Hoarder's race losses double), and the right of cardless players to keep declaring means the elimination the hoard defends against is only elimination from the asking half of the game. What this does *not* settle: the trigger competes against the roster's tuned ask policies, the systematic re-tuning protocol has never been run (Section 9), and the second containment mechanism — holding contained sets by default in the *declare* policy, which the opportunity-rate column suggests is where the leverage actually lives — is untouched in v0.5 [23].

9 Threats to validity

Stated as designed-for or open, per the laboratory's own checklist [24].

1. **Style as handicap.** Designed against: one shared full-strength inference engine; distinctness verified at the decision level. *Open:* the specified skill ablation — re-running every style over medium-strength inference to detect style-times-skill interactions — has not been run for this roster.

2. **Self-play convention lock-in.** A style that wins this tournament may be a style its own roster has learned to read [11]. *Open:* the holdout roster and the cross-play protocol against an independently written bot are specified but not yet run; the artifact's cross-play block is empty.

3. **Intransitivity misread as noise.** Designed against: explicit cyclic-energy metric, BH-corrected cycle enumeration, and a verdict rule with a first-class cyclic outcome.

4. **Tie-blindness.** Moot under `us54`: ties are impossible (Theorem 1) and the artifact asserts zero across 309,600 games.

5. **Passive-style stalling.** The step cap is instrumented, the stall-breaker is global (a per-style stall rule would be a hidden style parameter), and `cappedGames` = 0 in the reported run.

6. **Determinism mistaken for sampling.** Re-running a seed reproduces the game byte-for-byte; the harness asserts 4,300 distinct seeds.

7. **Multiplicity.** All 36 cells and all cycle tests are BH-corrected.

8. **Conditional on the rule set.** Every conclusion is conditional on `us54` as pinned by the artifact's rules hash. The sign-flip analysis is itself evidence that results do not transfer across dialects; nothing here is claimed for the 48-card game.

9. **An untuned roster.** The styles are evaluated *as specified*: the SPRT-based re-tuning protocol (each style tuned against the control) has never been run, so every per-style result — including both negative results — is about the specified vector, not the best version of the style that exists. A roster with weaker ask policies would, in particular, leave more room for the turn-pass of Section 8.3.

10. **Uniform teams only.** All cells are pure teams of three. The mixed-composition tier — including the hypothesis that the Archivist is weak in mirrors but strong as a single seat — is specified and unexplored; the artifact's teams block is empty.

10 Conclusion

Under the `us54` dialect, with style isolated from skill and the payoff matrix decomposed rather than assumed transitive, the question "is there a superior play style?" has a measured answer: yes — at this roster, at this level of play, the completion-chasing Punter dominates by all four pre-stated criteria, the matrix carries almost no cyclic energy, and the anticipated counter-structure is absent. The two results that qualify the headline are as load-bearing as the headline itself: the declare-threshold axis the styles are named after is inert across the roster's entire range, because the shared engine's confidence estimates are bimodal; and the hoarding thesis fails end-to-end, its theorized benefit surviving implementation only to be measured at zero. Honest instrumentation did not merely decorate these findings — the decision-divergence instrument caught a control masquerading as a style, and the paired v1/v2 re-run kept a nothing-result from being narrated into a something-result.

The natural next question is dynamic rather than static: whether an agent that identifies its opponents' styles online and adapts its own can beat the fixed Punter. That question is now answered, in the negative and with a measured cost, by the FishAI v1.0 companion paper [20] — distinct from the two companion papers on the inert axis and the contained book published alongside this one [18, 19].

Reproducibility. The engine is deterministic; the committed artifact [26] carries the seed set, per-cell records digest, engine commit and rules hash; and every number in this paper is either in that artifact or in the repository's measurement documents [22, 23, 24, 25].

11 Addendum: matrix v2, re-measured

Everything above reports one artifact, and that artifact has since been regenerated in place [27]. This section records what moved. It amends nothing: the figures in Sections 6–8 were correct readings of the artifact that existed when they were written, and they are left as they stand. That is the same discipline Section 8.3 applies to v1 — a superseded measurement is evidence about the engine that produced it, and a silent swap would destroy the very comparison this section is making.

11.1 What was re-run, and the confound

The re-measurement is controlled rather than an independent sample: the same 36 cells, the same 4,300 duplicate deal pairs per cell, the same `style-v1` seed prefix, 309,600 games, the same rules hash, and health gates clean again (`illegalActions`, `cappedGames`, `invariantViolations`, `ties`, `voids`, `nonClinch` all zero; 4,300/4,300 distinct seeds). Two things changed underneath it, and they changed *together*.

1. **The roster.** The concession layer [28] postdates engine commit `1667a1d`, so the vectors measured for this paper did not carry `defuse`. Every roster style now carries `defuse` = 1. That is a change to the games themselves, so it moves every cell of the matrix, not only the best-response search.

2. **The exploitability search.** Its knob ladder ran to 75 rungs and its candidate budget was a literal 60. The ladder now runs to 88 rungs — it gained `containedPass` (Section 8.3) and the concession layer's `defuse` and `conceal` — and the budget is the ladder's own length [29].

Every magnitude reported below is the joint effect of those two changes, and none of it may be attributed to the ladder alone. What *is* cleanly attributable to the ladder is a statement about reachability rather than about size, and it is in Section 11.4.

The budget deserves one paragraph of its own, because "60 of 75" overstates the damage and this paper should not bank an exaggeration in its own favour. A budget below the ladder's length does not sample the ladder — it evaluates a *prefix* and abandons the tail, which is a prune selected by the order the fields were typed rather than by evidence. But budget is spent only on candidates that actually change the incumbent, and 14–19 of the old ladder's 75 rungs were the incumbent's own value for any given style, so six of the nine targets finished the ladder inside 60. The artifact records `candidatesTried` of exactly 60 — the ceiling, not a count — for Banker, Turtle and Hoarder; replaying the old ladder along each style's committed acceptances reproduces all nine counts exactly and needs 61 for Banker and 62 for Turtle. Hoarder reached the last rung on its 60th spend and lost nothing. **Three rungs were lost in total: `minHandSize` 3 for Banker, and `minHandSize` 1 and 3 for Turtle** [29]. A small hole, in a badly chosen place — but a small one.

11.2 The verdict survives

All four criteria of Section 5.4 hold on the re-measured artifact and the computed verdict is again *dominant*: Punter still has the highest mean score, still has no losing matchup, the matrix is still almost purely transitive, and criterion 4 still passes. Table 13 puts the two columns side by side.

Criteria 1–3 survive with room to spare and with their structure intact. Punter's margin over Blitz halves but stays outside either run's interval; the maximin's worst-opponent label flips from Balanced to Blitz, which is the same relabelling between two near-tied cells that Section 6 recorded between v1 and v2 (the two cells are now 0.5112 and 0.5102, and all eight of Punter's cells remain

Table 13: The four criteria, as published (matrix v2 at engine commit `1667a1d`) and as re-measured (matrix v2 at `1fdd22e`) [26, 27]. The verdict is *dominant* in both.

#	Criterion	As published	Re-measured
1	highest mean score	Punter 0.5829, ahead of Blitz 0.5602 (margin 0.0227)	Punter 0.5684, ahead of Blitz 0.5562 (margin 0.0122)
2	maximin > 0.5	0.5190, worst vs. Balanced; CI_{95} lower bound 0.5119	0.5102, worst vs. Blitz; CI_{95} lower bound 0.5022
3	cyclic energy < 0.15	0.0112; one 3-cycle in point estimates, none significant	0.0172; one 3-cycle in point estimates, none significant
4	exploitability not materially worse	$E(\text{Punter}) = 0.0025$ vs. rivals' median 0.0444	$E(\text{Punter}) = 0.0525$ vs. rivals' median 0.0594

Table 14: Per-style mean score as published and as re-measured [26, 27]. Δ is a difference of point estimates between two runs on the same seed set; no paired per-deal statistic is quoted (see text).

Style	Published	Re-measured	Δ	Rank
Punter	0.5829	0.5684	−0.0145	$1 \to 1$
Blitz	0.5602	0.5562	−0.0040	$2 \to 2$
Balanced	0.5581	0.5531	−0.0050	$3 \to 3$
Banker	0.5285	0.4928	−0.0357	$4 \to 6$
Ghost	0.4928	0.5082	+0.0154	$5 \to 4$
Hoarder	0.4885	0.4897	+0.0012	$6 \to 7$
Archivist	0.4802	0.5023	+0.0221	$7 \to 5$
Scout	0.4063	0.4173	+0.0110	$8 \to 8$
Turtle	0.4025	0.4120	+0.0095	$9 \to 9$

above 0.5 and BH-significant); cyclic energy rises by half but is still an order of magnitude under its threshold, and 33 of 36 cells are significant, as before. Criterion 4 is the exception, and Section 11.4 is about it.

11.3 Every headline figure moved, and the ranking's interior re-ordered

Table 14 gives the per-style mean scores. The podium and the floor are unchanged — Punter, Blitz, Balanced at the top, Scout and Turtle at the bottom, in that order — but the middle of the table is not: Banker falls from 4th to 6th, Archivist rises from 7th to 5th, Ghost from 5th to 4th, Hoarder from 6th to 7th. The differences are stated as point estimates between two runs; each run's own 95% interval has a half-width near 0.003, and the paired per-deal test that Table 12 reports for the v1/v2 pair has not been run for this pair, so no t is quoted here. On that reading Banker's −0.0357 and Archivist's +0.0221 are the two large movements and the rest are small.

Two readings are available and this paper is not entitled to choose between them: a roster in which everybody defuses is a different tournament, and a re-ordering of the middle of a transitive ladder is what a real policy change to every seat should be expected to produce. Which of the concession layer's terms did it is a question for the document that ships them [28], not for a re-run of this one.

11.4 Criterion 4 is materially weaker

This is the finding of the addendum. This paper reports Punter's exploitability as 0.0025 against a rivals' median of 0.0444 — an eighteen-fold advantage, and the sentence in Section 6.5 that "it is

Table 15: Exploitability as published and as re-measured, ascending in the new E [26, 27]. **Bold** responses use knobs no earlier ladder contained.

Style	E (pub.)	Response (pub.)	E (new)	Response (new)
Hoarder	0.0563	`hoardBooks` 0, `minHandSize` 0	0.0000	`minHitP` 0.45, `leakEpsilon` 0
Banker	0.0012	`gambleBonus` 15	0.0212	`wHit` 60, `minHitP` 0.45
Archivist	0.0325	`wHit` 100	0.0463	`wNarrow` 25, `gambleBonus` 30
Punter	0.0025	`missTarget random`	**0.0525**	**`conceal` 1**
Blitz	0.0175	`leakEpsilon` 3	0.0587	**`defuse` 2, `conceal` 4**
Ghost	0.0775	`wProgress` 30, `gambleBonus` 30	0.0600	`wProgress` 45
Balanced	0.0000	none (mirror)	0.0613	**`conceal` 1**
Turtle	0.1500	`declareOnlyOwnHand` off	0.1325	`declareOnlyOwnHand` off
Scout	0.1112	`wHit` 90, `wNarrow` 0, `minHitP` 0.45	0.1475	`wHit` 80, `wNarrow` 0, `missTarget most`

not a paper champion of this particular roster" leans on the size of that gap. On the re-measured artifact $E(\text{Punter}) = 0.0525$ against a rivals' median of 0.0594. That is essentially a tie. Punter is now the *fourth* least exploitable of the nine (Table 15), where it was third and within 0.0025 of the floor. Criterion 4 passes on the letter of the rule — "not materially worse", with the lab's margin of 0.0200 — and on nothing more. That is a far weaker statement than this paper's prose around it implies, and readers should take the prose, not the pass, as the thing that has been corrected.

The reachability argument, which is clean. The search that produced 0.0025 ran over a ladder that contained no rung for `containedPass`, none for `defuse`, and none for `conceal`: no committed exploitability number in this paper has ever priced any of those three mechanisms [29]. Three of the nine new best responses are drawn from exactly there — `conceal` 1 against Balanced, `conceal` 1 against *Punter itself*, and `defuse` 2 with `conceal` 4 against Blitz. Those three deviations were not merely missed by the old search; they were unreachable by it *by construction*. The control makes the point most sharply: the old search spent 56 candidates against Balanced and accepted nothing at all, reporting a flat 0.0000; the new one accepts a single knob worth 0.0613.

The confound, which is equally real. The roster also changed between the two runs. The old Es were maxima over a search against opponents that did not defuse, computed on a matrix whose every cell has since moved (Table 14). So the *magnitudes* — $0.0025 \to 0.0525$, and the median's $0.0444 \to 0.0594$ — are joint effects, and this paper does not claim the ladder caused them. The honest split is: the ladder is why three particular exploits exist at all, and the two changes together are why the numbers are what they are.

Why this is a result to own rather than a correction to bury. Section 6.5 already states, as one of its two honest cautions, that E is a maximum over a stochastic search — and Section 5.4's criterion is a comparison of such maxima. A maximum over a search is a *lower* bound on exploitability, so it can only rise as the search grows, and the size of the gap between two styles' Es is a property of the ladder as much as of the styles. This addendum is a concrete instance of that caveat firing: enlarge the search by three axes, and the dominant style's eighteen-fold advantage becomes a tie. The claim criterion 4 is entitled to make is "no cheap single-knob counter to Punter was found by this search", and the re-measurement is a reminder of how much work "this search" does in that sentence.

Two further cautions carry over unchanged and one is new. The final evaluations still carry standard errors of 0.012–0.017, so differences between adjacent rows of Table 15 are not meaningful — Punter's 0.0525, Blitz's 0.0587, Ghost's 0.0600 and Balanced's 0.0613 are four numbers inside one standard error of each other, and Punter's fourth place should be read as "in the middle of the pack", not as a rank. The search still only accepts moves worth about 0.03 or more. And new:

the Hoarder's 0.0000 is a clamp, not a measurement of unexploitability — its accepted candidate scored 0.54125 in search and 0.480 ± 0.0125 on the held-out block, below the mirror baseline, and $E = \max(0, \mathrm{SR} - 0.5)$ [27].

11.5 What the re-measurement does not disturb

Caveat I (Section 7) is a decision-level measurement made with the divergence instrument on replayed games, not a reading of the payoff artifact, and the re-measurement does not bear on it. Caveat II's central claim — that the contained-book turn-pass is worth nothing — rests on the mirror measurement of Table 11 and the paired v1/v2 comparison of Table 12. Both remain valid, and for a reason worth stating precisely: that comparison isolates the turn-pass because v1 and the artifact this paper reports differ by the policy change and nothing else. The re-measured artifact differs from v1 by the turn-pass *and* the concession layer, so it cannot be substituted into Table 12, and any re-derivation of that pairing from it would be measuring two changes and attributing them to one. The one part of Caveat II that does not survive is the exploitability sharpening, recorded where it appears in Section 8.3.

Finally, the threats of Section 9 are unmoved, and this addendum adds nothing to the list so much as it supplies the first worked example for two entries already on it — "an untuned roster" and the standing caution that every result here is conditional on the engine that produced it.

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